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Electronics 1 Lab (ECE 2410)

School of Sustainable Systems
and Engineering

Laboratory Experiment (1)

Diode Characteristic

1. Objective:

To explore the characteristics of ordinary diodes through the use of mathematical modelling and implement circuit testing.

Upon completion of this laboratory experiment, you should have a good understanding of the electrical characteristics and the parameters affecting the design of semiconductor junction diodes.

1. Theory:

A diode is a semiconductor device that conducts current much more readily in one direction than in the other. Its familiar shape is shown in Fig. 1.1:

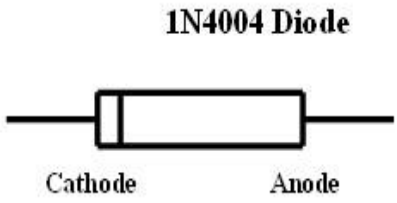


Fig. 1.1

The standardized symbol (IRC 60617, ANSI/IEEE 315A-1986) is shown in Fig. 1.2a but all along you will find variations like in Fig. 1.2b.

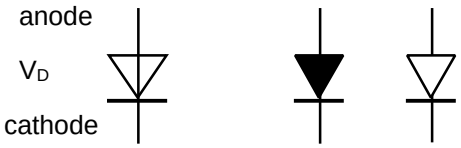


Fig. 1.2a

Fig. 1.2b

When the diode's anode is at higher potential than the cathode, the diode is "forward biased" by the "forward voltage" V_F. The "forward current" I_F will flow from the anode to the cathode in the direction the symbol's arrowhead (see also the definition of the polarities of I_F and V_F in Fig. 1.3).

Unlike a resistor in which the current is proportional to the voltage drop, the diode is a nonlinear device. When the diode is forward biased a small voltage drop (often called the threshold voltage V_{th}) occurs across the diode (for germanium V_{th} ~ 0.3 V, silicium V_{th} ~ 0.7 V).

This relationship between the voltage across diode and the current passing through it can be seen in the following detailed characteristic curve diagram, Fig. 1.3.

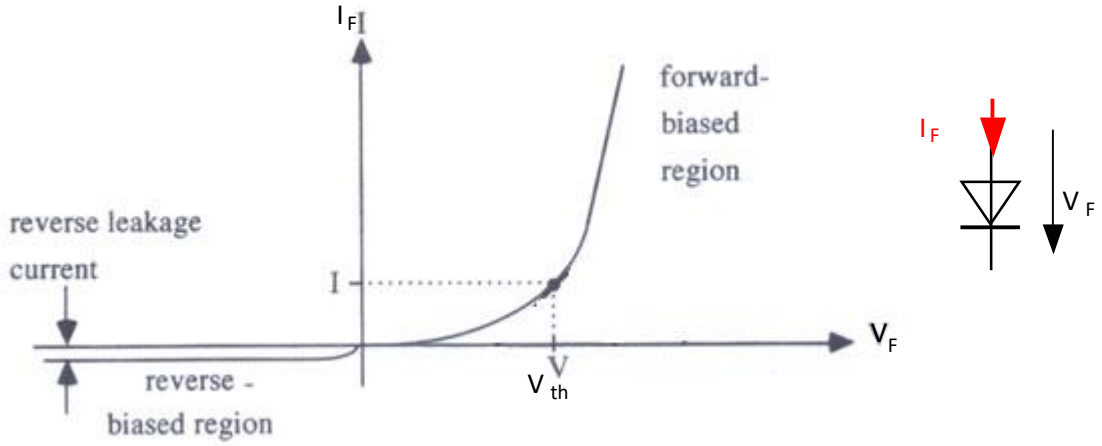


Fig. 1.3

3. *Equipment & Instruments*

Oscilloscope
Function Generator
Multimeter
Module no. Dl3155E12
Breadboard
Resistors and Diodes

3. *Pre-lab*

1. Use your favourite circuit simulation software and simulate the circuit in Fig 1.5. Set the input voltage to the values: 0.5V, 0.7V, 1V, 5V, and record the diode voltage and current in each case.
2. Simulate the circuit in Part II and record the voltages V_1 and V_o with the diode and without the diode.
3. Write a short pre-lab report with results.

4. *Procedure*

Part I: Diode Characteristics

1. Check the functionality of the Silicon Diode D1 using the DMM.
2. Make a wired connection between +V from the base frame Power Supply to input terminal 1 and ground of the power supply terminal to the ground of the circuit in block no. 4.
3. Turn the knob +V on the base frame power supply to fully counter-clockwise.
4. Vary the voltage V_R (using knob +V) from V_R = 0.1 V to V_R = 0.9 V in steps of ΔV_R = 0.1 V, and from V_R = 1 V to V_R = 10 V in steps of ΔV_R = 1 V, in each step measure the diode voltage V_F and the diode current I_F.
   Record the data in tables like Tab.1.1, Tab.1.2 and Tab.1.3.
5. Sketch $I_F = f(V_F)$; describe the nature of the curve. Note that we are using the diode in the forward bias mode.

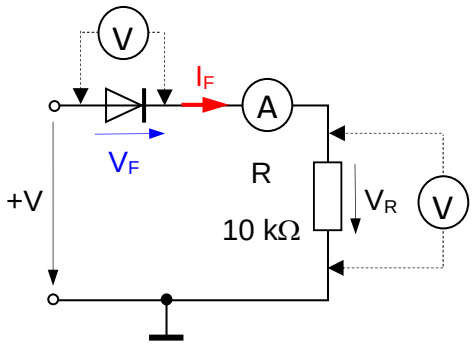


Fig. 1.5

V_s /V	0.1	0.2	0.3	0.4	0.44	0.46	0.48	0.50	0.52	0.54
V_F /V										
I_F / mA										

Tab. 1.1

V_s /V	0.56	0.58	0.60	0.62	0.64	0.70	0.75	0.80	0.85	0.90
V_F /V										
I_F / mA										

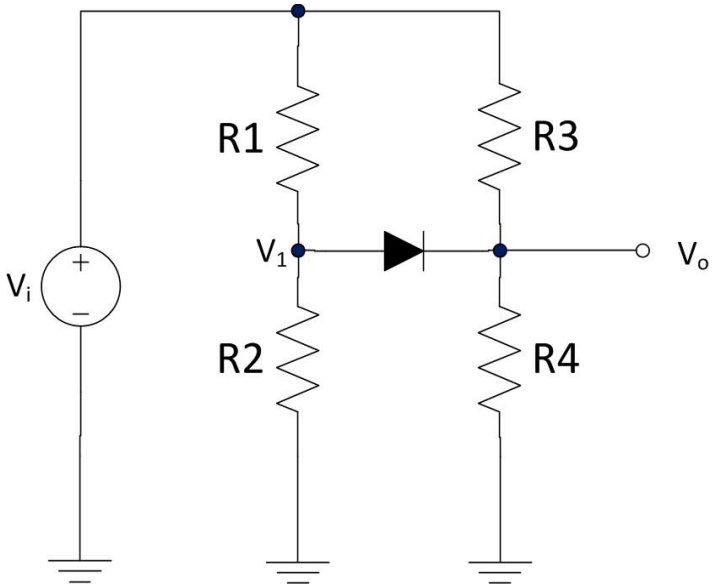
Tab. 1.2

Vs /V	1	2	3	4	5	6	7	8	9	10
V_F /V										
I_F / mA										

Tab. 1.3

Part II: Diode Circuits

1. Build the following circuit on a breadboard with R1=1kΩ, R2=R3=R4=2kΩ and V_i=10V DC.



2. Measure the voltages V_1 and V_o.
3. Remove the diode and measure the voltage V_1 and V_o.
4. Explain the difference in the voltages measured in steps 2 and 3.

Part III: Diode Circuit Design

1. Given only 1 resistor, diodes and capacitors, design a circuit that outputs 1.2V DC (+/- 5%) when the input is 5V DC.
2. Implement and test the circuit.